Connected to Stay: Gender Homophily and Its Role in Open-Source Software Developer Retention

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Introduction

Open-source software (OSS) communities are essential infrastructures for global digital collaboration, yet sustaining participation remains difficult. Persistent gender imbalances—where women account for less than 5% of active contributors—compound challenges of equity and long-term engagement. Beyond technical skills, social connections play a critical role in retention, influencing whether contributors remain active or disengage.

This study investigates *gender homophily*—the tendency of individuals to collaborate with others of the same gender—as a structural factor shaping persistence in OSS. Homophily may provide social support for underrepresented contributors, yet excessive gender segregation could reinforce inequality. We explore how gender-based collaboration patterns form, persist, and predict contributor retention.

The central research questions are: (1) To what extent do OSS networks display gender homophily beyond structural expectations? and (2) How does homophily influence the likelihood of continued participation, particularly for women contributors? We conceptualize OSS ecosystems as dynamic social systems where network structure and inclusion interact to determine community sustainability.

Methods

This study combines large-scale network analysis and longitudinal modeling using data from *World of Code (WoC)*, covering over one million developers across 14 years (2008–2022). OSS collaboration networks were built annually, where nodes represent contributors and edges represent co-participation in projects. Edge weights corresponded to shared project frequency. Developers were disambiguated via identity resolution algorithms, and gender was inferred probabilistically using cross-cultural name models. Analyses were limited to contributors with identifiable gender while preserving overall network structure.

To quantify gender homophily, observed-to-expected ratios compared same-gender and cross-gender collaboration frequencies against randomized null models preserving network degree distributions. Ratios greater than one indicated overrepresentation of same-gender ties. Homophily strength was measured across direct and indirect paths to capture both local and systemic effects.

Inferential modeling used *Exponential Random Graph Models (ERGMs)* to test whether same-gender ties occur beyond what network topology would predict. Models estimated the conditional probability of forming same-gender collaborations while controlling for network density, triadic closure, and prior exposure to women collaborators. Parameters were estimated via Monte Carlo maximum likelihood with post-stratification corrections for gender imbalance.

To connect social structure to persistence, developer retention was modeled using *Cox proportional hazards regression*. “Departure” was defined as the absence of commits in subsequent years. The model estimated how having at least one woman collaborator affected dropout risk, controlling for collaboration degree and project activity. Analyses were stratified by gender to identify differential effects. Lower hazard ratios indicated stronger retention associated with diverse collaborations.

This multi-method design integrates macro-level structural analysis and micro-level behavioral modeling. It provides both descriptive and inferential insights into how social connectivity—and its gendered dimensions—predicts sustained participation in digital ecosystems.

Potential Impact

The findings reveal that gender homophily has a measurable impact on contributor retention. Developers with gender-diverse collaboration networks were significantly less likely to disengage, indicating that inclusivity strengthens community resilience. For women in particular, access to supportive ties mitigates isolation in male-dominated environments.

By modeling OSS as an evolving social system, this work shifts the focus from individual dropout to structural resilience. Inclusive collaboration patterns are shown to diffuse stability across networks, suggesting that diversity is a collective asset rather than a demographic goal alone.

The implications extend to both research and practice. For OSS platforms, interventions such as mentorship matching, recognition of collaborative labor, and network-aware recommendations can strategically reinforce inclusive structures. Theoretically, the study contributes to computational sociology by linking gender equity to systemic sustainability. In demonstrating how social similarity and diversity co-produce resilience, it highlights inclusion as a measurable and actionable dimension of digital ecosystem health.